

Small Corporate Network — Packet Tracer Project

A simulated small corporate network built in Cisco Packet Tracer, covering VLAN segmentation, inter-VLAN routing, DHCP, dual-WAN redundancy, firewall NAT/ACLs, and wireless security.

Topology Overview

- **Core:** 3560-24PS Multilayer Switch (inter-VLAN routing)
- **Access switches (5x 2960-24TT):** Server Room, Guest Network, IT Department, Corporate Network, Office Users
- **Firewall:** Cisco ASA 5506-X (NAT, guest isolation ACL)
- **WAN:** Dual 1941 routers (primary + backup) for redundancy
- **Wireless:** Corporate Wi-Fi (WPA2-PSK) and Guest Wi-Fi APs

VLANs

VLAN	Name	Subnet
10	IT	10.10.10.0/24
20	Office	10.10.20.0/24
30	Servers	10.10.30.0/24
40	Corp_WiFi	10.10.40.0/24
50	Guest_WiFi	10.10.50.0/24
60	Voice	10.10.60.0/24

Files

- `Small_cooperate_network.pkt` — the Packet Tracer project file
- `docs/VLAN_Trunk_Config_Rebuild.md` — VLAN/trunk configuration for all switches
- `docs/Full_Network_Buildout.md` — IP addressing, routing, DHCP, WAN, firewall, and wireless configuration

Highlights / Lessons Learned

- Rebuilt VLAN and trunk configuration from a clean slate across 6 switches
- Resolved native VLAN mismatches between trunk links
- Implemented inter-VLAN routing via SVIs on the core switch
- Configured DHCP relay (`ip helper-address`) across VLANs to a centralized DHCP server
- Set up dual-WAN routing with a primary and backup path through the firewall
- Applied a guest network isolation ACL on the ASA firewall
- Upgraded wireless security from WEP to WPA2-PSK